ModelIt!

Student Activity Pack

# Gene Regulatory Networks

From Transcription Factor to Phenotype — Modeling the Logic of Gene Expression

Grade 09L3 | NGSS HS-LS3-1, HS-LS1-4 | Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**1 If every cell in your body has the same DNA, why is a brain cell so different from a skin cell?**

**2 What do you think controls which genes are turned on or off in a particular cell at a particular time?**

**3 Draw a diagram showing how you think a cell 'reads' a gene and makes the protein it codes for — include as many steps as you can.**

**4 What do you think happens when a gene that should be turned off gets stuck in the 'on' position?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

- Transcription Factor Concentration
- Promoter Binding Affinity
- mRNA Transcription Rate
- mRNA Degradation Rate
- Ribosome Availability
- Protein Translation Rate
- Protein Degradation Rate
- Feedback Signal Strength
- Epigenetic Modification
- Regulatory Network Stability

| EXTERNAL | INTERNAL |
|---|---|
| | |

### Explain your reasoning: Why did you sort each component the way you did?

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Steady-State Expression

Moderate transcription factor, open chromatin, negative feedback

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Positive Feedback Switch

Strong positive autoregulation

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Epigenetic Silencing

Closed chromatin, high transcription factor

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**MicroRNA Regulation** — Small non-coding RNA molecules (18-25 nucleotides) that bind to complementary sequences in mRNA and either block translation or trigger mRNA degradation — adding a powerful post-transcriptional layer of gene regulation that affects over 60% of human genes

**Chromatin Remodeling Activity** — The activity of ATP-dependent chromatin remodeling complexes (SWI/SNF, ISWI, CHD, INO80) that physically reposition nucleosomes to expose or hide gene promoters — the molecular machines that execute epigenetic decisions about gene accessibility

**Stochastic Noise Level** — The inherent randomness in gene expression caused by the probabilistic nature of molecular interactions — when key regulatory molecules are present in low copy numbers (10-100 per cell), random fluctuations can cause genetically identical cells to express genes at dramatically different levels

## My Extension

### A Which new component would you add to the model?

### B How would it connect to the existing components? (Draw or describe)

### C What evidence or reasoning supports your connection?

# 7 Reflection Questions

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### The Big Question

**Your body has 20,000 genes but every cell only uses a fraction of them. Who decides which genes turn on, which stay off, and when — and what happens when that decision-making system breaks?** Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that gene expression is not a simple on-off switch but a dynamic balance of production and degradation at every level?**

2. **What is the functional difference between negative feedback and positive feedback in gene regulatory circuits — and what happens to Regulatory Network Stability in each case?**

3. **Why does Epigenetic Modification act as a master override that can silence a gene regardless of Transcription Factor Concentration?**

4. **How would adding Stochastic Noise Level to your model change predictions about cell-to-cell variation in gene expression?**

5. **What are the medical implications when gene regulatory networks malfunction — and how does your model help explain diseases like cancer?**

# 8 STEM Challenge: Design a Synthetic Gene Circuit for Biosensing

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## YOUR ENGINEERING MISSION

Design a synthetic gene regulatory circuit that detects an environmental signal (toxin, pathogen, pollutant) and produces a visible output (fluorescent protein, color change, electronic signal) for use as a living biosensor.

### THE SCENARIO

A water quality monitoring agency needs a low-cost, deployable biosensor that can detect heavy metal contamination (lead, arsenic, mercury) in drinking water supplies across rural communities where laboratory testing is unavailable. Your synthetic biology team must design a gene regulatory circuit that senses the presence of heavy metals and produces a visible fluorescent signal proportional to contamination level. The biosensor must be sensitive, specific, and stable.

### Engineering Design Thinking

1. HOW WOULD YOU ENSURE THE BIOSENSOR REMAINS FUNCTIONAL IN VARIABLE ENVIRONMENTAL CONDITIONS (TEMPERATURE, PH, SALINITY)?

2. WHAT OUTPUT SIGNAL WOULD BE MOST PRACTICAL FOR FIELD DEPLOYMENT — FLUORESCENCE, COLOR CHANGE, OR ELECTRONIC SIGNAL — AND WHY?

3. HOW WOULD YOU CALIBRATE THE BIOSENSOR SO THAT FLUORESCENCE INTENSITY CAN BE CONVERTED TO A QUANTITATIVE CONTAMINATION READING?

4. WHAT BIOCONTAINMENT STRATEGY WOULD YOU USE TO PREVENT THE ENGINEERED BACTERIA FROM ESCAPING INTO THE WATER SUPPLY BEING TESTED?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Systems Biologists and Synthetic Biologists design and analyze gene regulatory networks for applications in medicine, biosensing, and biomanufacturing. They work for biotech companies (Ginkgo Bioworks, Twist Bioscience), pharmaceutical companies, academic research labs, and government agencies, earning $85,000-$180,000/year. Bioinformaticians who computationally model gene regulatory networks earn $80,000-$160,000/year.